

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:

Cloud Computing
and Big Data
Analytics

COURSE CODE:

CSA15

89
100

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)

Duration: 60 Minutes
25

Total Marks:

Question Count: 5

Code of Conduct C192511296)

I A. Subashini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Subashini

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

90/

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CSA1521
cloud computing

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Evolution from standalone computing to cloud computing

1. Standalone computing

- * One computer works by itself.

- * All software and data are stored on that - computer.

- * No Internet needed.

eg:- using MS word on your personal computer without connecting to any network.

2. Client - server computing

- * Many computers (clients) connect to one central server.

- * The server stores data and provides service.

eg: college students access marks from the college server

3. Distributed computing

- * Many computers work together and share tasks.

- * work is divided to improve speed.

eg: several computers processing different parts of the same project

4. Grid computing

- * computers in different locations work together as one large system
- * used for very large scientific calculations.
- Eg: weather forecasting and scientific research.

5. Virtualization

- * one physical computer is divided into many virtual machines (VMs)
 - * Each VM acts like a separate computer.
- Benefit: Better use of hardware and lower cost.

6. Cloud computing

- * computer resources (servers, storage, software) are provided over the Internet.
- * users can access them anytime and pay only for what they use.

Technologies that Enabled cloud computing.

- * High speed Internet - connect users to cloud services
- * Virtualization - Run multiple virtual computers on a single physical machine
- * Data centers - large buildings with thousands of servers
- * Distributed computing - shares work among many computers
- * APIs and web services - Allow different cloud applications to communicate

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Hypervisors in virtualization

A Hypervisor is software that creates and manages Virtual Machines (VMs).

Type 1 allows multiple operating systems (Windows, Linux, Ubuntu, etc.) to run on a single physical computer at the same time.

Example: on computer can run Windows and Ubuntu simultaneously using a Hypervisor.

Type 1 Hypervisor (Bare-Metal)

* Installed directly on the hardware.

* No host operating system is required.

* Faster and more secure.

* Mostly used in data centers and cloud computing.

Eg:-

* VMware ESXi

* Microsoft Hyper-V * Xen

Type 2 Hypervisor (Hosted)

- * Installed on top of a existing operating system.

- * Depends on the host OS.

- * Easier to install but slightly slower.

- * Mostly used for testing, learning and personal use.

Example:

- * Oracle VirtualBox

- * VMware Workstation.

API

Application programming interface.

is a set of rules that allows different software applications on cloud services to communicate and exchange data.

API = messenger.

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* API send a success message back to the app

4) Resource pooling

Resource pooling means the cloud providers combines resources such as CPU, memory, storage and network into a shared pool. These resources are allocated to users whenever they need them.

Working

- * Many users share the same cloud infrastructure.
- * Each user gets the required resource automatically.
- * When a user finishes, the resources are returned to the pool and can be used by others.
- * Users do not know the exact physical location of the resources.

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APIs support Interoperability

Interoperability means different applications on cloud platforms can work together even if they are from different providers.

APIs make this possible by:

- * Using standard communication

Methods (such as REST or HTTP).

- * Allowing secure data exchange.

- * Enabling different cloud services to communicate without compatibility issue.

Eg:

A mobile app uploads photos to Google Drive using the Google Drive API.

Flow:

- * user uploads a photo
- * app sends request through API
- * Google Drive stores photo

5) Rapid elasticity

Rapid elasticity is the ability of a cloud system to quickly increase or decrease resources (CPU, RAM, storage, or decrease servers) automatically based on user demand.

How it works

- * when more users access an application, the cloud adds more resources automatically.
- * when fewer users are using it, the cloud removes the extra resources.
- * This happens quickly without interrupting the service.

Eg: Imagine an online shopping website during festival sale:

- * Normal day - 1,000 users $\rightarrow$ 2 servers are enough

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Benefits to Multiple users:

- * Efficient resource utilization:
no resources are wasted.
- * Cost-effective: users pay only for what they use.
- * Scalable: Resources can be increased or decreased based on demand.
- * High-Availability: Resources are always available when needed.
- * Multi-tenancy: Many users can safely share the same infrastructure.

eg:-

- * Many students use the same book.
- * Each student borrows a book when needed and returns it after use.
- * The same books are then available for other students.

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Festival sale: 100,000 users → cloud automatically increases to 20 servers.

* After the sale: user traffic drops → cloud reduces back to 2 servers.

Rapid Elasticity helps by:

* Automatically increase resource during high Demand.

* Reducing resource when demand is low.

* Maintaining fast performance.

* Avoiding server overload.

* Reducing cost by using only the required resources.